

Implementation of Linear Controllers

Rotary Inverted Pendulum

Project Report by

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Abstract

The inverted pendulum problem is a classic control problem with broad applications in aerospace, robotics, and various fields of control engineering. It serves as a benchmark for testing and studying advanced control strategies. A rotary inverted pendulum is a highly non-linear, unstable, and underactuated mechanical system. In this work, insights are provided into deriving the system's mathematical model. The report presents the stabilization of a Quanser Rotary Inverted Pendulum at its upright position using Pole Placement, linear quadratic regulator (LQR), and gain-scheduled LQR techniques.

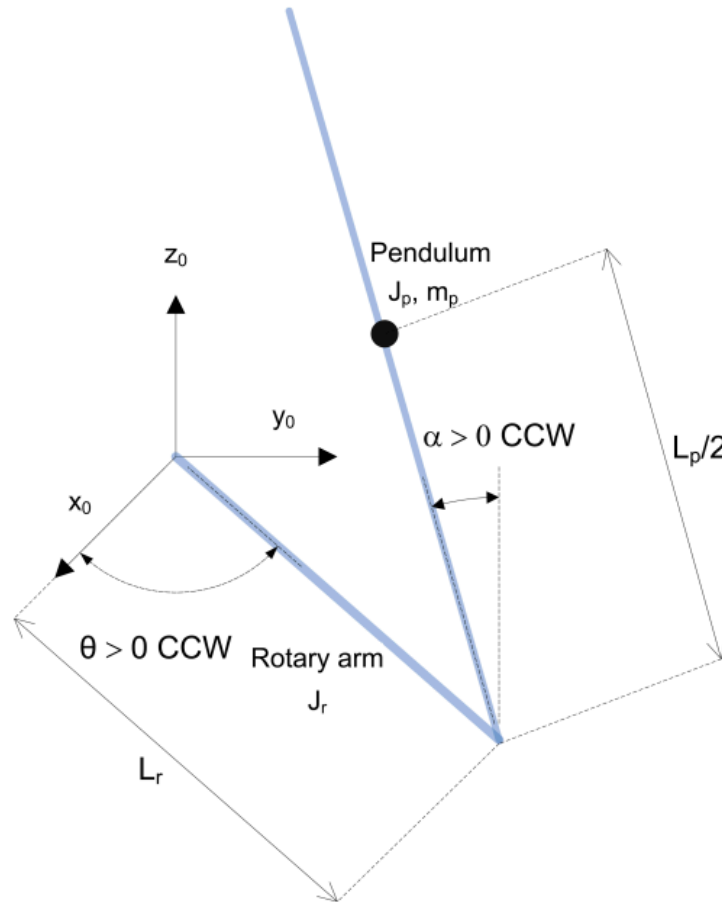
The linearized system model was validated from various online references, and the controllers were tested both in simulation and on the hardware setup. Experimental results demonstrate that the system achieved a steady-state error of $\pm 4^\circ$ with a settling time of 6 seconds for a 20° setpoint using LQR. The Gain-Scheduled LQR controller successfully stabilized the pendulum across a sweep range of $\pm 45^\circ$ at a cost of performance. Future work will focus on developing a swing-up controller to automate the transition from the pendant to the upright position.

Keywords

Rotary Inverted Pendulum, Underactuated System, Euler-Lagrange Modeling, Pole Placement, Linear Quadratic Regulator (LQR), Gain Scheduling, Linearization, Quanser setup.

Introduction

The inverted pendulum is a standard control problem widely used as a first-principles reference for many complex, coupled nonlinear systems. For every control engineer, it lies as the foundation and also establishes the basis for modern control theory and discrete recursive systems. Although the system has only two angles to change and its angular velocities, it is capable of providing only a single input, namely motor voltage (SIMO system).



The rotary arm has a length L_r and a moment of inertia J_r . Its angular position, θ , is defined to be positive when the arm rotates in the counterclockwise (CCW) direction. The servo motor, and therefore the arm, produces CCW motion when the applied motor voltage is positive ($V_m > 0$).

A pendulum link is attached at the distal end of the rotary arm. The pendulum has a total length L_p , and its center of mass is located at a distance $L_p/2$ from the pivot. Its moment of inertia about the center of mass is J_p . The pendulum angle, α , is defined as zero when the pendulum is perfectly vertical in the upright equilibrium configuration. A positive α corresponds to a CCW rotation away from the upright position.

The derivation of the Rotary Inverted Pendulum is straightforward; a summarized approach is provided [here](#).

After taking the eigenvalues of the system, one of the four eigenvalues is at zero, and another is on the positive real axis, indicating that the system is inherently unstable.

Designing of Controller

Pole placement simulation:

```
zeta = 0.707;  
wn = 5;  
real_part = -0.7 * 4;  
imag_part = wn * sqrt(1 - zeta^2);  
des_poles = [real_part + imag_part*i, real_part - imag_part*i, -30, -40];
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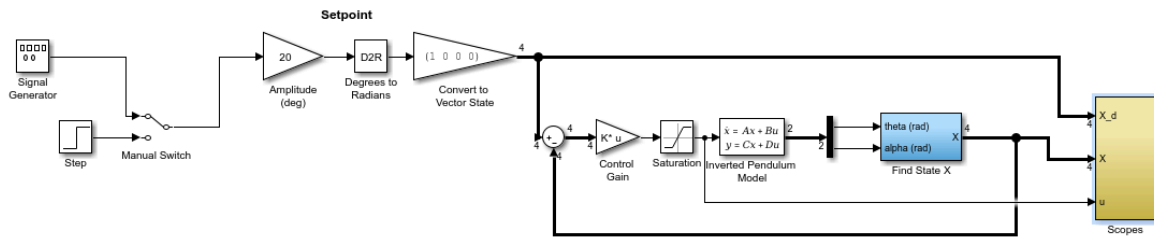
As some of the poles(or eigenvalues here) in the system make it inherently unstable, the poles are moved to the left plane to make it stable. The dynamics of the system depend on the dominant poles.

The two dominant poles are placed to be conjugate pairs, as an underdamped system has a faster rise time compared to other cases.

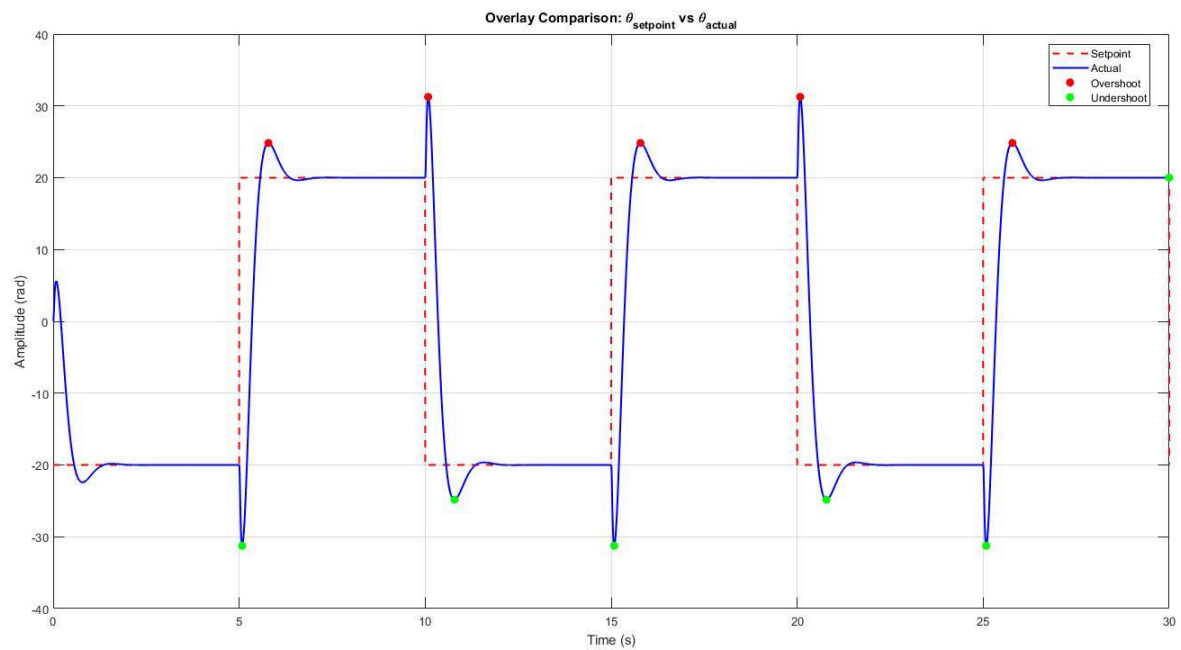
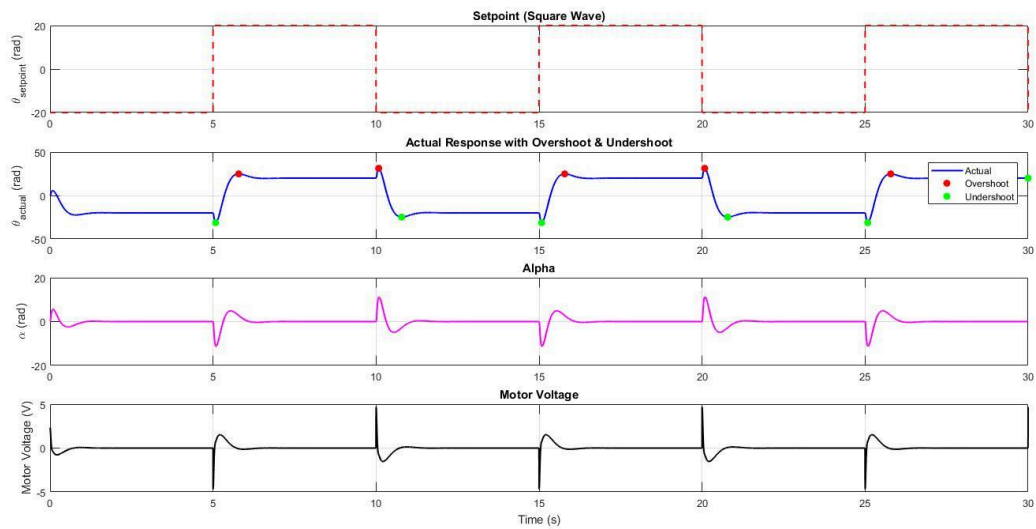
We had chosen a settling time of 1.132s, approximating it as a second-order system when it is actually a fourth-order system, by considering the effect that the other two eigenvalues are not dominant. With a general thumb rule, we take the remaining poles to be -30 and -40. Evidently, as the order of the system increases, it is far from being approximated to a second order, which means that the settling time would never be 1.132 for this rotary inverted pendulum.

For the chosen poles, the values and graphs correspond to their output.

Idx	Type	Overshoot (%)	Undershoot (%)	Peak Time (s)	Peak Value	Min Time (s)	Min Value
1	Rise	12.109	28.15	5.786	24.843	5.08	-31.259
2	Fall	128.15	12.109	10.08	31.259	10.786	-24.843
3	Rise	12.109	28.15	15.786	24.843	15.08	-31.259
4	Fall	128.15	12.109	20.08	31.259	20.786	-24.843
5	Rise	12.109	28.15	25.786	24.843	25.08	-31.259



System Response: Setpoint, Actual, Alpha, and Motor Voltage



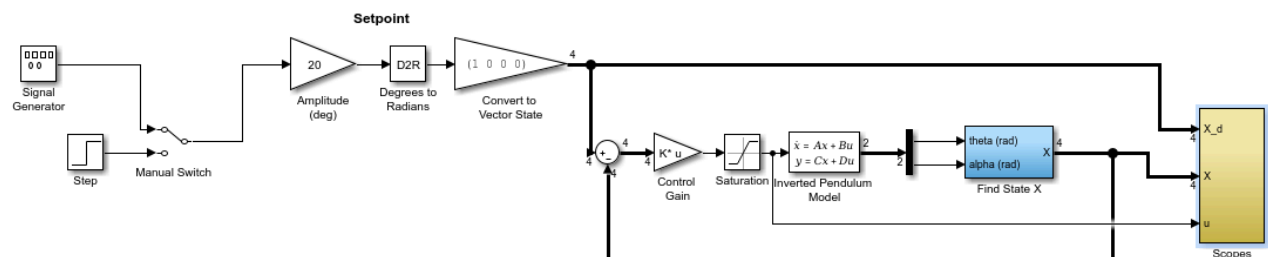
LQR simulation:

The linear quadratic regulator is an abstracted version of pole placement, where, instead of working directly with the eigenvalues, we use two different weight matrices, Q and R, such that the squared cost function of the states and input is minimized. Due to this abstract math involving weights, finding the global minima depends on determining the appropriate Q and R matrices through trial and error; otherwise, it could lead to a local minima, rather than the global minima. There are several methods, like Bryson's method, to find an initial Q and R matrix to start with.

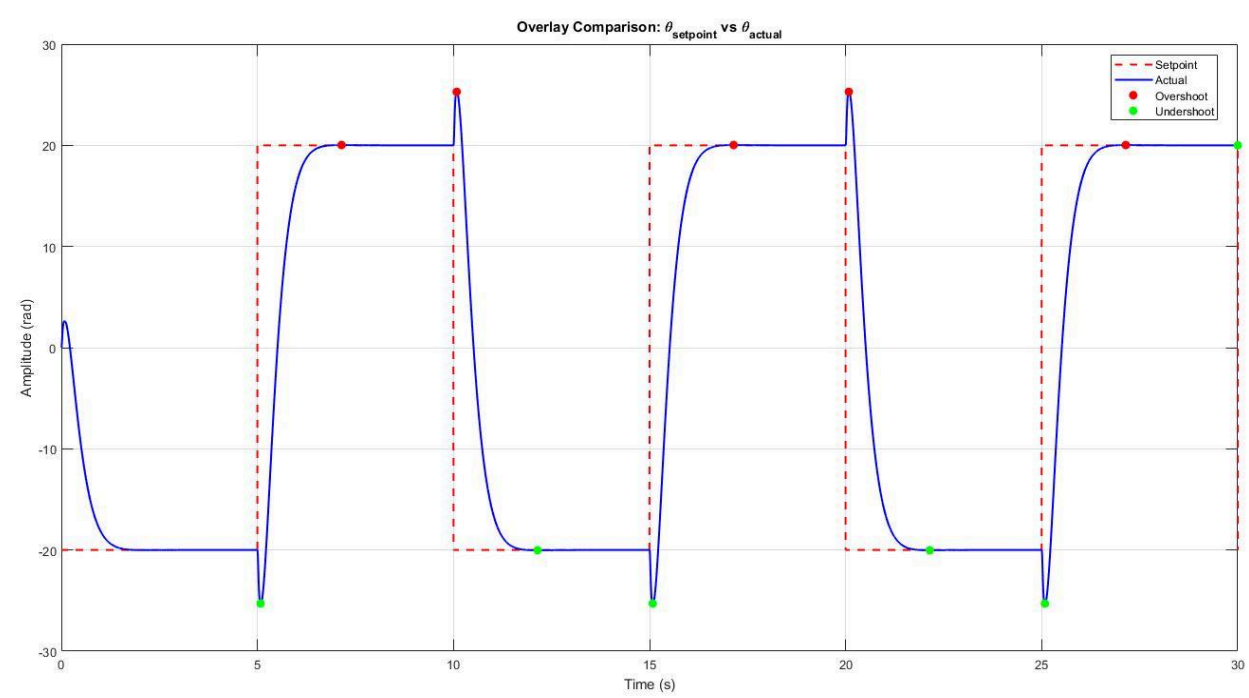
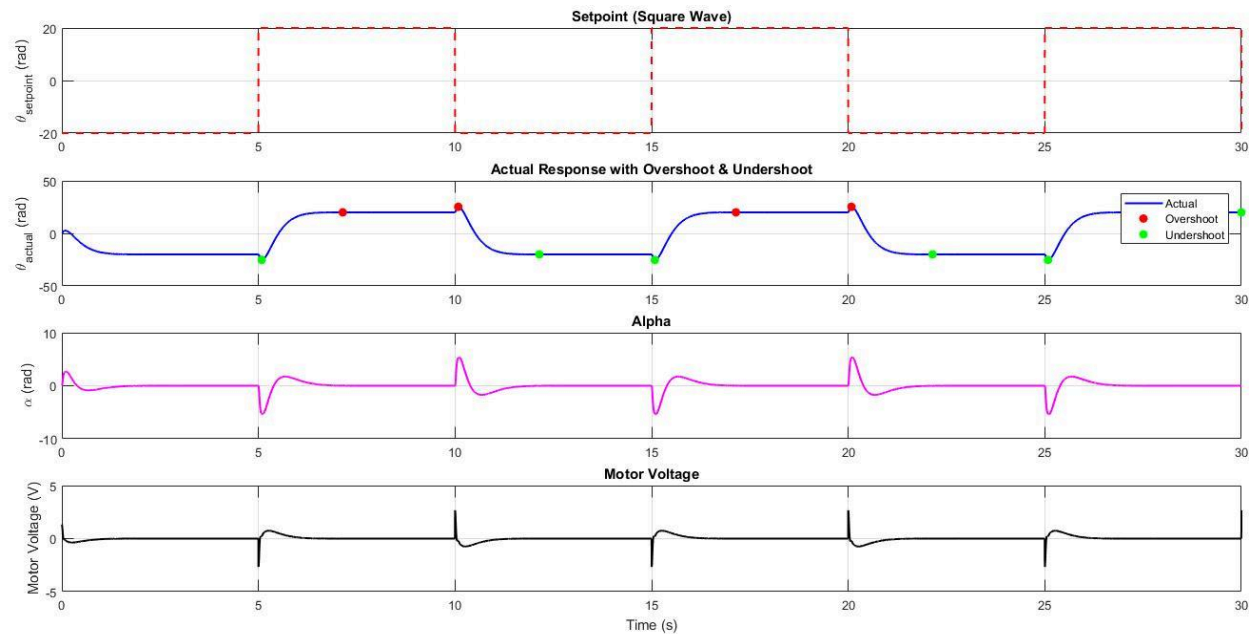
As you could see from the results below, LQR stands superior compared to pole placement due to its faster settling time and minimal overshoot.

Idx	Type	Overshoot (%)	Undershoot (%)	Peak Time (s)	Peak Value	Min Time (s)	Min Value
1	Rise	0.064984	13.23	7.144	20.026	5.082	-25.29
2	Fall	113.23	0.064984	10.082	25.29	12.144	-20.026
3	Rise	0.064984	13.23	17.144	20.026	15.082	-25.29
4	Fall	113.23	0.064984	20.082	25.29	22.144	-20.026
5	Rise	0.064984	13.23	27.144	20.026	25.082	-25.29

SRV02 Inverted Pendulum: Balance Control



System Response: Setpoint, Actual, Alpha, and Motor Voltage

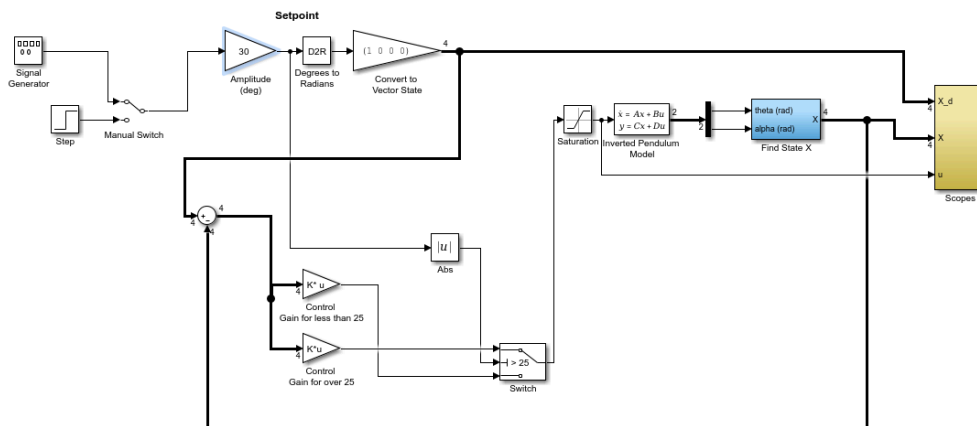


LQR Gain Scheduling:

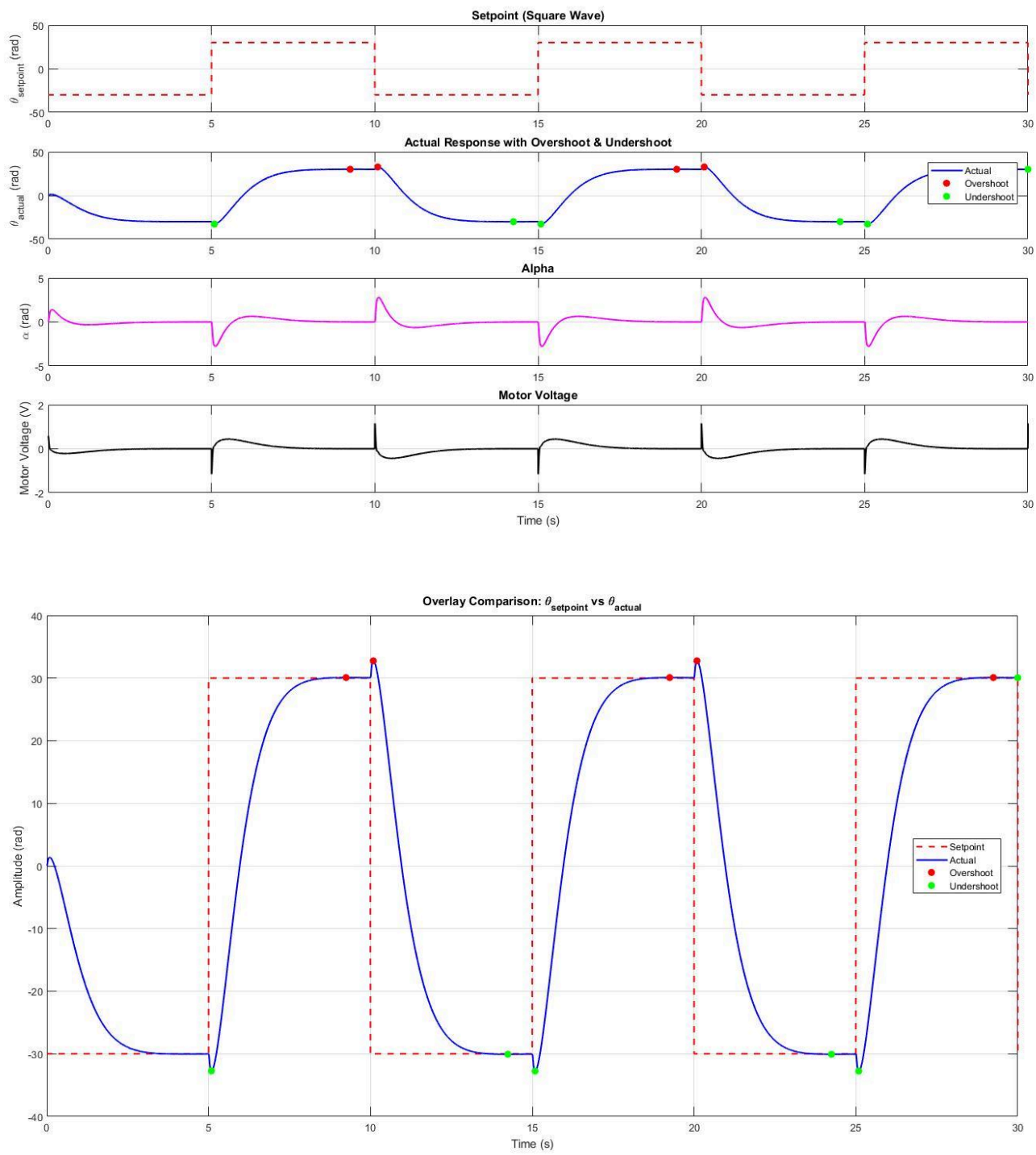
This part is entirely optional for simulation but essential for hardware implementation. When the setpoint is increased to higher values, the α will grow so much that it exceeds the linearized range (from the first-order linearization using Taylor series), resulting in different behavior that was not accounted for in the model. A weaker controller is used for higher θ rates of change in alpha, as the rate of change in alpha is reduced compared to the previous aggressive controller.

Idx	Type	Overshoot (%)	Undershoot (%)	Peak Time (s)	Peak Value	Min Time (s)	Min Value
1	Rise	0.039193	4.62	9.246	30.07	5.088	-32.724
2	Fall	104.65	0.039214	10.088	32.743	14.246	-30.07
3	Rise	0.039214	4.65	19.246	30.07	15.088	-32.743
4	Fall	104.65	0.039214	20.088	32.743	24.246	-30.07
5	Rise	0.039214	4.65	29.246	30.07	25.088	-32.743
6	Fall	0.000000	0.000000	30.000	30.041	30.000	30.041

SRV02 Inverted Pendulum: Balance Control



System Response: Setpoint, Actual, Alpha, and Motor Voltage



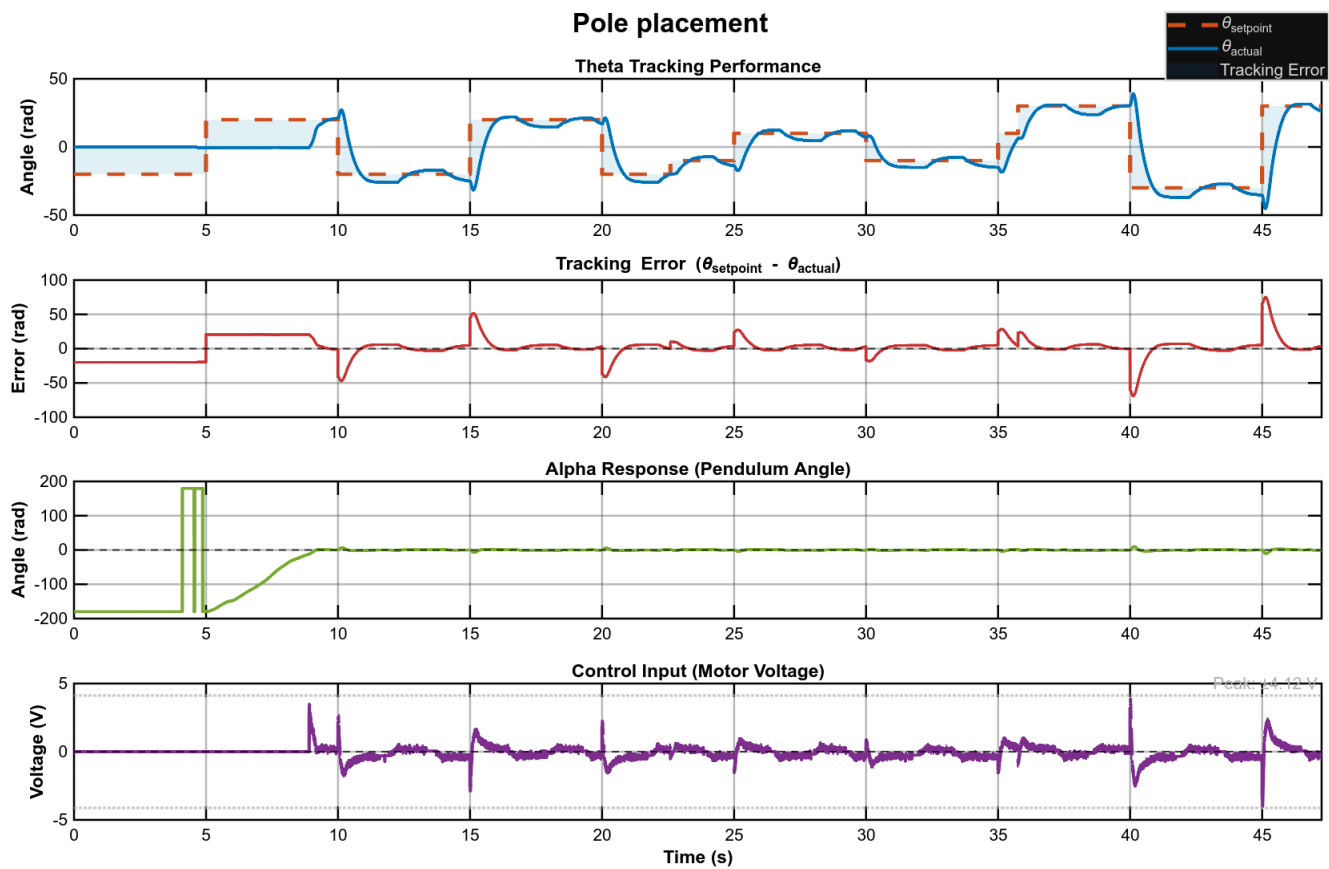
Hardware implementation

The beauty of Control Systems is that, regardless of the controller you design, there is always a case where it fails in reality. Below is the given hardware setup of Quanser's Rotary inverted pendulum. The hardware in the loop is interfaced by Quanser's proprietary input ports and its power module with the base computer.

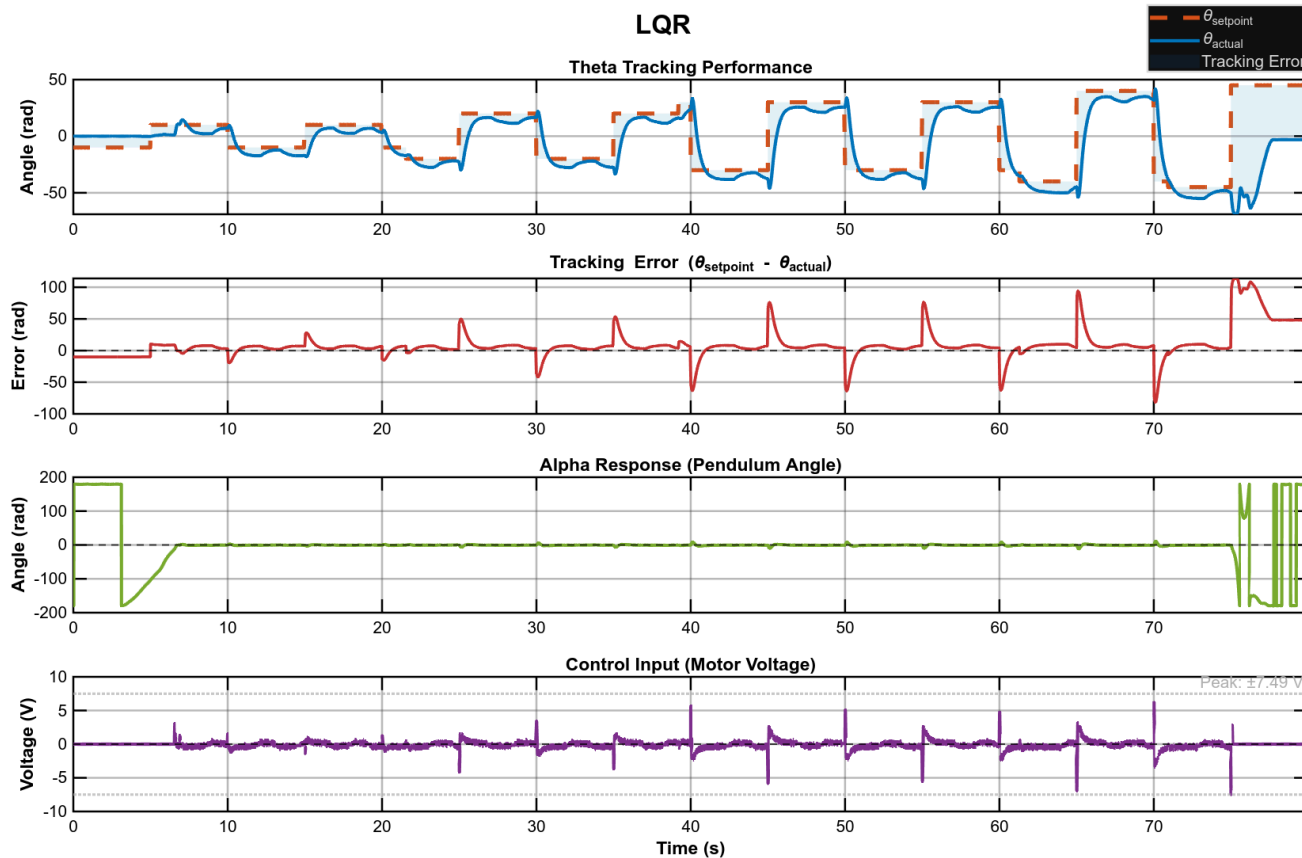
In the given graphs, the controller is designed to start only after the pendulum is almost vertical. That's why, for some initial time, when the pendulum is raised, the theta is flat, and the alpha slowly increases from -180° to 0° . So ignore the initial values until the alpha is closer to zero.



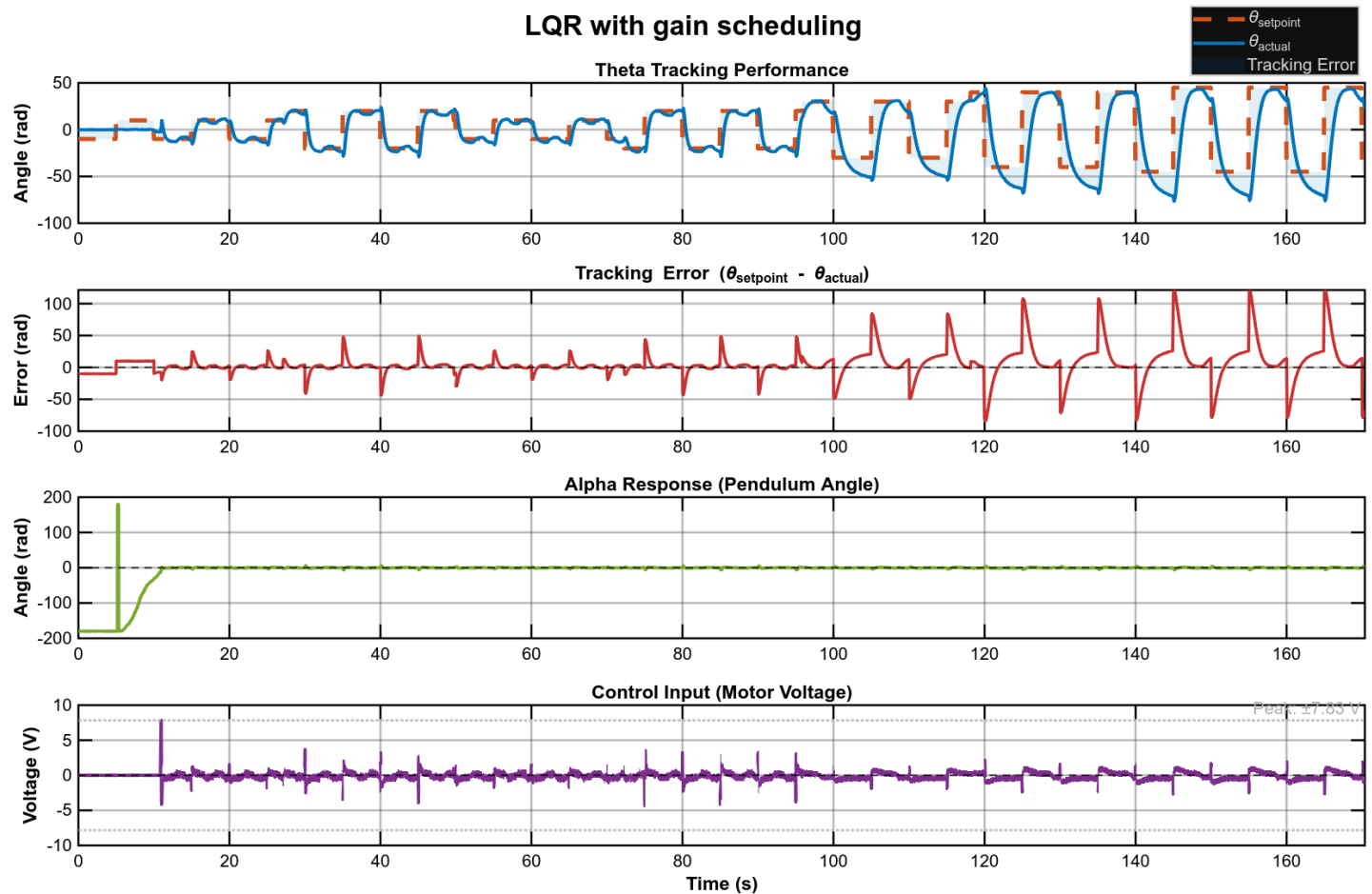
Pole placement results:



LQR:



LQR with gain scheduling:



From the results, for lower θ values, LQR and Pole placement work almost similarly. The system achieved a steady-state error of $\pm 4^\circ$ with a settling time of 6 seconds for a 20° setpoint using LQR, and for higher θ values, both pole placement and LQR fail. Gain-scheduled LQR is stable for higher θ values and can sweep a range of $+45^\circ$ to -45° , but at a cost of tracking the setpoint.

Video references:

Pole placement: [link](#)

LQR: [link](#)

LQR with gain scheduling: [link](#)